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|---|---|---|---|---|---|
| 5 | (a) | (i) | displacement is the distance the rope / particles are (above or below) from the equilibrium / mean / rest / undisturbed position (not 'distance moved') | B1 | [1] |
| | | (ii) | 1. amplitude (= 80 / 4) = 20 mm | B1 | [1] |
| | | | 2. $v = f\lambda$ or $v = \lambda / T$ | C1 | |
| | | | $f = 1 / T = 1 / 0.2$ (5 Hz) | C1 | |
| | | | $v = 5 \times 1.5 = 7.5\ \mathrm{m\,s^{-1}}$ | A1 | [3] |
| | (b) | | point A of rope shown at equilibrium position | B1 | |
| | | | same wavelength, shape, peaks / wave moved ¼$\lambda$ to right | B1 | [2] |
| | (c) | (i) | progressive as energy OR peaks OR troughs is/are transferred/moved /propagated (by the waves) | B1 | [1] |
| | | (ii) | transverse as particles/rope movement is perpendicular to direction of travel /propagation of the energy/wave velocity | B1 | [1] |